

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I Avishek N(192511009)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Avishek N

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

ARTICLE SELECTION AND REVIEW BASIS

The assessment requires one peer-reviewed research article published after 2019 and related to Inductive Learning, Decision Trees, Statistical Learning Methods, or Reinforcement Learning applications. The selected article directly addresses the CO4 topic of Decision Trees through a healthcare classification problem.

Selected article: Ilyas, H., Ali, S., Ponum, M., Hasan, O., Mahmood, M. T., Iftikhar, M., and Malik, M. H. "Chronic kidney disease diagnosis using decision tree algorithms." BMC Nephrology, Volume 22, Article 273, 2021. DOI: 10.1186/s12882-021-02474-z.

CONTENTS

- 1. Task 1 – Article Summary
- 2. Task 2 – Methodology Analysis
- 3. Task 3 – Results and Critical Evaluation
- 4. Task 4 – Reflection and Learning Outcome
- 5. Integrated Discussion
- 6. Conclusion
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TASK 1 – ARTICLE SUMMARY

1(a) Full Citation, Domain and ML Problem

Full citation: Ilyas, Hamida; Ali, Sajid; Ponum, Mahvish; Hasan, Osman; Mahmood, Muhammad Tahir; Iftikhar, Mehwish; and Malik, Mubasher Hussain. "Chronic kidney disease diagnosis using decision tree algorithms." BMC Nephrology, 22, 273 (2021). DOI: 10.1186/s12882-021-02474-z.

Domain/Application Area: Artificial Intelligence and Machine Learning applied to healthcare and medical decision support. The application is chronic kidney disease diagnosis and prediction of CKD severity stages.

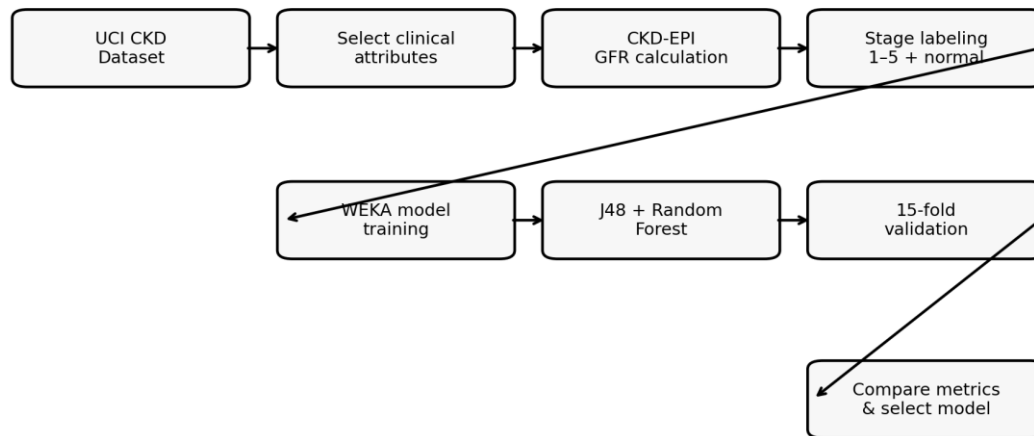
Specific ML problem: The paper formulates CKD-stage prediction as a supervised multiclass classification problem. The UCI dataset contains 400 instances and 25 attributes. The original classes are CKD and NOTCKD; the authors further subdivide the CKD condition into stages based on estimated GFR: Stage 1, Stage 2, Stage 3A, Stage 3B, Stage 4 and Stage 5.

1(b) Key Objective, Research Gap and Proposed Solution

The central objective is to investigate whether machine-learning classification can support earlier and more detailed identification of chronic kidney disease severity. The authors transform clinical records into a multiclass prediction problem and use the CKD-EPI equation to estimate GFR from age, sex, race and serum creatinine. J48 and Random Forest classifiers are then used in WEKA to predict CKD stages. The models are evaluated using 15-fold cross-validation and metrics including accuracy, sensitivity, specificity, precision, F-measure and execution time. The reported overall results are 85.5% accuracy for J48 in 0.03 seconds and 78.25% for Random Forest in 0.28 seconds.

Research gap: The authors state that earlier studies had explored several ML approaches for CKD prediction, but they identify a specific gap in predicting CKD stages using age, sex, race and serum creatinine through GFR-based staging. Their contribution is therefore stage-wise multiclass classification using two tree-based methods and a common validation framework.

Proposed Methodology Flow



TASK 2 – METHODOLOGY ANALYSIS

2(a) Machine Learning Techniques Used

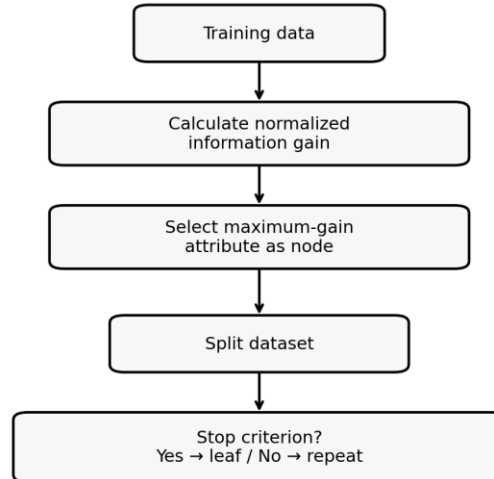
J48 Decision Tree

J48 is the WEKA implementation of the C4.5 decision-tree algorithm. It is an inductive supervised-learning method in which labelled training examples are used to construct a hierarchy of decision rules. Internal nodes represent attribute tests and leaf nodes represent predicted classes. J48 can handle numerical and categorical data and uses normalized information gain to choose useful splitting attributes.

Random Forest

Random Forest is an ensemble supervised-classification method. It creates many decision trees from bootstrapped samples and random subsets of variables. For a new observation, the trees generate predictions and majority voting determines the final class. The article describes repeated tree construction and the use of out-of-bag instances. The authors also note that Random Forest is computationally more expensive and less straightforward than a single J48 tree.

J48 (C4.5) Decision-Tree Learning Flow



Dataset Description

The dataset is obtained from the University of California Irvine (UCI) Machine Learning Repository and contains 400 instances and 25 attributes. The dataset initially has two classes: CKD and NOTCKD. The study creates a detailed stage representation by using calculated GFR values. Age, sex, race and serum creatinine are selected as inputs for the CKD-EPI GFR calculation.

Data preparation is organized into preprocessing, computation and final-result phases. Relevant attributes are selected, GFR is estimated, stage labels are established and the resulting data are classified using WEKA. Testing and validation are performed using 15-fold cross-validation.

GFR as a Derived Variable

Glomerular filtration rate is used as the basis for kidney-function severity. The paper compares CKD-EPI with MDRD and chooses CKD-EPI because it is considered more precise for estimating GFR across the relevant stages. The GFR calculation creates the bridge between clinical measurements and the final multiclass stage labels.

Hardware and Software

The reported hardware consists of an Intel Core i5 CPU at 2.40 GHz, 4 GB RAM and a 64-bit operating system. The computational engine is implemented using WEKA.

2(b) MAPPING TO CO4 TOPICS

CO4 Topic	Use in Article	Evidence	Alignment
Inductive Learning	Learns general rules from labelled CKD examples.	Training data → information gain → tree → prediction.	Direct
Decision Trees	J48/C4.5 is the principal model.	Maximum normalized information gain selects splits.	Core topic
Statistical Learning	Uses several classification metrics.	Accuracy, sensitivity, specificity, precision, F-measure.	Strong support
Reinforcement Learning	Not used in this article.	No reward-driven agent/environment loop.	Contrast only

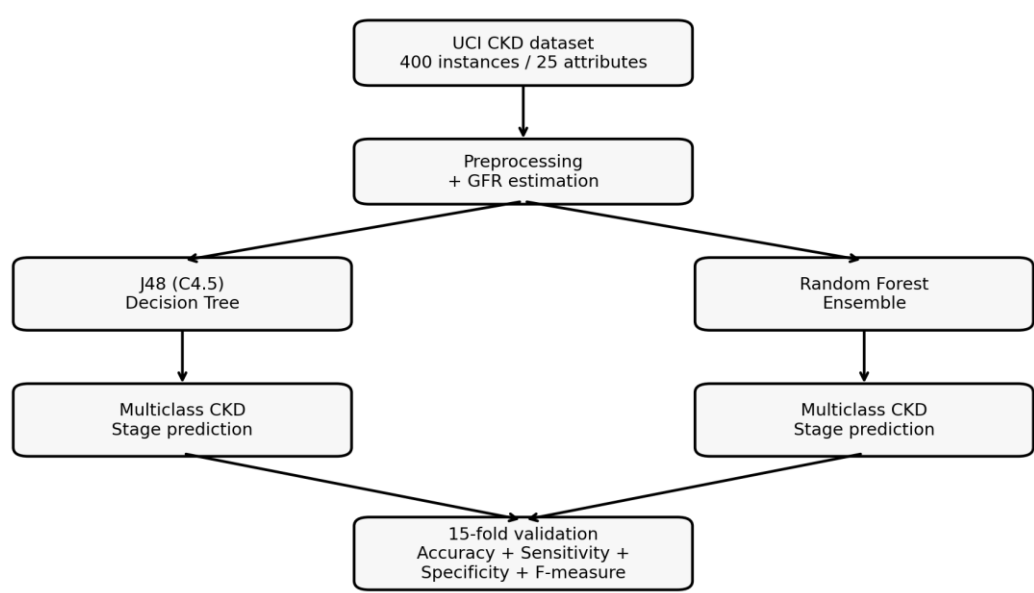
Methodological Strength

A major strength is the direct comparison of a single interpretable decision tree with an ensemble tree method under the same application setting. J48 permits a human-readable decision path, while 15-fold cross-validation provides a systematic estimate of predictive performance. The study also reports several complementary metrics instead of accuracy alone.

Methodological Limitation

The main limitation is the relatively small dataset of only 400 instances. A single-source dataset may not represent the diversity of real hospital populations, laboratory practices or disease distributions. The paper itself acknowledges that a larger dataset could improve performance. The study also does not provide external validation on an independent clinical population.

CO4 Decision-Tree Article Review: Algorithmic Pipeline



TASK 3 – RESULTS AND CRITICAL EVALUATION

3(a) Key Results

The study evaluates six CKD severity categories. Across the individual stages, J48 generally provides stronger classification results than Random Forest. The most important overall comparison is 85.5% overall accuracy for J48 with 0.03 seconds execution time versus 78.25% for Random Forest with 0.28 seconds.

CKD Stage	J48 Accuracy	RF Accuracy	J48 F-measure	RF F-measure
Stage 1	96%	96%	0.55	0.250
Stage 2	96%	93%	0.71	0.449
Stage 3A	98%	98%	0.77	0.64
Stage 3B	94%	93%	0.81	0.757
Stage 4	95%	87%	0.88	0.746
Stage 5	93%	91%	0.66	0.561
Overall	85.5%	78.25%	—	—

The stage-wise values are taken from the authors' reported summaries. Stage 4 shows a particularly large difference in accuracy: 95% for J48 versus 87% for Random Forest. Stage 3A has 98% accuracy for both methods, while J48 reports stronger sensitivity and F-measure.

3(b) Critical Evaluation

Are the metrics realistic? The reported results are plausible for a controlled benchmark dataset, but they should not be interpreted as clinical-grade diagnostic accuracy. An overall accuracy of 85.5% is meaningful as an experimental result, yet healthcare decisions require attention to the consequences of false negatives and false positives.

Potential dataset bias: Only 400 instances are available, and they originate from one public repository. This may limit representation of different demographic groups, hospitals, laboratory practices and disease prevalences. The stage labels also depend on GFR estimation, so errors or assumptions in this derived variable can influence the classification target.

Potential evaluation bias: Random shuffling followed by 15-fold cross-validation is useful, but all folds still come from the same underlying dataset. Similar patterns may therefore appear in training and testing folds. Independent external validation is necessary before claiming broad generalization.

Metric interpretation: Multiple metrics are essential. For example, Stage 1 J48 has 96% accuracy but only 56% sensitivity. This demonstrates that high overall accuracy can coexist with weaker detection of a particular class.

Additional Experiments

- External validation using a second hospital or independent CKD dataset.
- Repeated stratified cross-validation with confidence intervals.
- Class-imbalance analysis and macro-averaged precision, recall and F1-score.
- Hyperparameter tuning for Random Forest and J48.
- Comparison with Logistic Regression, SVM and Gradient Boosting.
- Feature-importance and explainability analysis.
- Prospective evaluation using newly collected patient records.

3(c) REAL-WORLD APPLICABILITY

A decision-tree-based CKD screening tool could be deployed as a clinical decision-support component rather than a replacement for physicians. A hospital information system could collect patient attributes, estimate GFR, classify likely CKD stage and display the decision path. Because J48 is relatively fast and interpretable, it is suitable where transparency and low computational overhead are important.

Industry-Scale Challenges

- Data availability and reliable labels.
- Interoperability between hospital information systems.
- Generalization across populations and institutions.
- Privacy, cybersecurity and medical-data governance.
- Clinical accountability and human oversight.
- Continuous monitoring for model and data drift.
- Fairness and unequal performance across demographic groups.

15-Fold Cross-Validation Concept

Dataset shuffled and divided into 15 folds



One fold is used for testing; the remaining 14 folds are used for training.

Repeat until every fold has served as the test fold → average performance.

TASK 4 – REFLECTION & LEARNING OUTCOME

4(a) Three Key Insights

Insight 1 – Decision Trees Produce Explainable Rules

The article strengthened my understanding that a decision tree is not only a classifier that outputs a label; it is also a structured representation of decision rules. J48 uses normalized information gain to select useful attributes and organize the tree. This directly connects to the CO4 syllabus topic of Decision Trees and explains why tree-based models are attractive when interpretability matters.

Insight 2 – Validation Is as Important as Model Selection

The use of 15-fold cross-validation demonstrates that an ML model should not be judged from a single training result. Each fold is used as a test set once while the remaining folds are used for training. This connects CO4 model-building knowledge with statistical evaluation and shows why reported accuracy must always be interpreted together with the validation protocol.

Insight 3 – Accuracy Alone Can Hide Class-Specific Weakness

The paper reports accuracy, sensitivity, specificity, precision, F-measure and ROC area. Stage 1 demonstrates why this matters: J48 achieves 96% accuracy but 56% sensitivity. Therefore, a high accuracy value does not guarantee reliable detection of every class. This connects directly to statistical learning and classification evaluation.

4(b) Novel Extension

I propose a cost-sensitive, externally validated CKD-stage prediction system using a larger multi-hospital dataset. J48 would remain the interpretable baseline, while tuned Random Forest and Gradient Boosting would be added as stronger ensemble baselines. Class-specific costs would penalize clinically important false negatives more heavily.

The extension is feasible because the original study already establishes the data-processing pipeline, GFR-based staging and WEKA tree-learning workflow. A larger dataset could follow the same feature definitions while adding stronger validation. The expected impact is improved generalization, more clinically meaningful evaluation and a clearer understanding of whether J48 remains competitive on unseen populations.

INTEGRATED DISCUSSION: CONNECTION TO CO4

The selected article provides a practical example of how CO4 concepts connect into a complete machine-learning workflow. The process begins with a real-world healthcare problem, acquires labelled data, prepares features, creates GFR-based stage labels, trains supervised classifiers, validates the models and compares performance using quantitative metrics. This is an example of inductive learning because the system learns general decision rules from labelled examples and applies them to unseen records.

Why J48 Is Central to the Review

J48 is directly relevant to the assignment because it is a decision-tree algorithm and an implementation of C4.5. Internal nodes represent attribute-based tests and leaves represent predicted classes. The model is comparatively interpretable, making it useful for explaining why a record is assigned to a particular category.

Why Random Forest Is a Useful Benchmark

Random Forest is built from multiple decision trees and aggregates their predictions. It provides a natural benchmark against J48 because both methods are tree-based but differ in complexity. The article's experiment shows that the more complex ensemble was not automatically superior: J48 achieved better overall accuracy and required less execution time in this dataset.

Meaning of the Overall Result

The reported 85.5% J48 accuracy should be understood as an experimental benchmark, not as a guarantee of safe diagnosis for future patients. The small dataset and absence of external validation limit generalization. The strongest conclusion is that the study demonstrates the feasibility of decision-tree-based CKD-stage classification and identifies J48 as the stronger method of the two under the tested conditions.

Practical Significance

The work demonstrates that useful ML systems require more than choosing an algorithm. Data preparation, feature engineering, validation, multiple metrics, interpretability, computational efficiency, privacy and clinical oversight all influence whether a model is suitable for deployment.

CONCLUSION

This article review examined the 2021 research article “Chronic kidney disease diagnosis using decision tree algorithms” by Ilyas et al. The paper addresses CKD severity-stage classification using the UCI CKD dataset, GFR-based staging, J48 and Random Forest. The study evaluates the models using 15-fold cross-validation and several classification metrics.

The reported results show that J48 achieved 85.5% overall accuracy in 0.03 seconds, compared with 78.25% accuracy in 0.28 seconds for Random Forest. J48 also performs better across most stage-level measures. The article therefore provides a strong CO4 example of inductive learning and decision-tree classification. At the same time, the relatively small dataset and lack of external validation limit how far the results can be generalized to real-world clinical populations.

Overall, the article demonstrates that responsible machine-learning evaluation requires attention to the data, algorithm, validation method, metrics and deployment environment. A larger multi-hospital study with external validation and cost-sensitive evaluation would be a strong next step toward real-world applicability.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article’s purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand